

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) designing and deploying agent-based agentic AI workflows, RAG systems, and multi-agent orchestration for enterprise production environments. Experienced translating complex AI capabilities into scalable, reliable solutions for non-technical stakeholders. Delivered ~35% retrieval precision gain, >30% hallucination reduction, and ~30% fewer post-deployment defects. Python proficiency; REST API integration and JSON payload handling for enterprise systems.

CORE COMPETENCIES

Agentic AI Workflow Design • RAG Systems & Memory Management • Multi-Agent Orchestration • Prompt Engineering • REST API Integration • LangChain / LlamaIndex • Python / FastAPI / Docker • Enterprise Solution Delivery • Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS

- Designed and deployed **agentic AI workflows** for multi-step healthcare queries using structured reasoning, tool discipline, and grounding rules; improved stability by ~25%, reduced hallucinations by >30%.
- Built **RAG systems** with context-aware LLM responses for clinical knowledge retrieval; ~35% precision gain, 90%+ grounding alignment in evaluation.
- Developed **advanced prompting and memory management** techniques for complex multi-step workflows.
- Packaged AI inference as **FastAPI/Flask services** with Docker; integrated with enterprise systems via REST APIs; reduced post-deploy defects ~30%.
- Collaborated cross-functionally to integrate AI workflows into existing healthcare IT frameworks.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- RAG-grounded document Q&A; reduced hallucinations ~30%, raised accuracy >85%. Flask API for non-technical users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- SQL/T-SQL optimization; REST API integration experience; Jenkins CI/CD deployment automation (>30% error reduction).

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Risk Scoring) with schema-validated agent contracts, RAG-grounded validation, advanced prompting, and audit-ready reasoning traces. Direct parallel to Workato's multi-agent collaboration and RAG pipeline design.

Dream Decoder - Multimodal AI Orchestration

FastAPI + React/TypeScript app orchestrating Perplexity Sonar API + Replicate image synthesis. Intermediate structured prompt transformation layers improved contextual alignment ~30% and success rate ~25-30%.

Manga Lens - Multi-Provider AI Integration

Production Chrome extension integrating four AI providers (Claude, GPT-4o, Ollama). Multi-provider payload abstraction, 7-day cache, 29 sites. REST API integration across multiple enterprise AI backends.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio

Aug 2023 - May 2025

SKILLS

Languages: Python, Java (basic), TypeScript/JavaScript, SQL

LLM & AI: RAG, agentic workflows, LangChain, LlamaIndex, prompt engineering, vector search, structured outputs

APIs & Integration: REST, JSON, FastAPI, Flask, Docker

ML: scikit-learn, XGBoost, PyTorch, HuggingFace

Work Authorization: F-1 OPT (US-authorized now; future sponsorship required)